

Linux File Management Machine Test

Objectives

- 1) Create an empty file called **x** in the **/tmp** directory.
- 2) Copy the **x** file from **/tmp** to your home directory.
- 3) Create a copy of the **x** file and name it **y**.
- 4) Create a directory called **files** and move both **x** and **y** into it.
- 5) Create a copy of the **files** directory and name the copy **copy_of_files**.
- 6) Rename the **copy_of_files** directory to **files2**.
- 7) Remove both **files** and **files2** directories along with their contents.

Output:----

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ cd /tmp  
sreelekha@LAPTOP-CKI0PC9R:/tmp$ touch x  
sreelekha@LAPTOP-CKI0PC9R:/tmp$ cd ~  
sreelekha@LAPTOP-CKI0PC9R:~$ cp /tmp/x .  
sreelekha@LAPTOP-CKI0PC9R:~$ ls  
AWS          Linux        backup       linux-test  
DevOps       LinuxPractice config.env   project  
DevOps-Team  MyLinuxLab  deploy.sh   project.txt  
DevOpsLab    Test        devops      x  
Docker       Ubuntu      employee.txt  
Kubernetes   application.log linux  
sreelekha@LAPTOP-CKI0PC9R:~$ cp x y  
sreelekha@LAPTOP-CKI0PC9R:~$ mkdir files  
sreelekha@LAPTOP-CKI0PC9R:~$ mv x y files/  
sreelekha@LAPTOP-CKI0PC9R:~$ ls files  
x  y  
sreelekha@LAPTOP-CKI0PC9R:~$ cp -r files copy_of_files  
sreelekha@LAPTOP-CKI0PC9R:~$ ls  
AWS          Linux        backup       files  
DevOps       LinuxPractice config.env   linux  
DevOps-Team  MyLinuxLab  copy_of_files linux-test  
DevOpsLab    Test        deploy.sh   project  
Docker       Ubuntu      devops      project.txt  
Kubernetes   application.log employee.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ mv copy_of_files files2  
sreelekha@LAPTOP-CKI0PC9R:~$ rm -r files files2  
sreelekha@LAPTOP-CKI0PC9R:~$
```

Linux File and Directory Management Machine Test

Objectives

1. Create a file called **x**.
2. Create a directory called **content**.
3. Move the **x** file to the **content** directory.
4. Create a file inside the **content** directory called **y**.
5. Create the following directory structure inside the **content** directory:
dir1/dir2/dir3.
6. Remove the **content** directory along with all its contents.

Output:--

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ touch x  
sreelekha@LAPTOP-CKI0PC9R:~$ mkdir Content  
mkdir: Content: File exists  
sreelekha@LAPTOP-CKI0PC9R:~$ mv x Content/  
sreelekha@LAPTOP-CKI0PC9R:~$ ls Content  
x  
sreelekha@LAPTOP-CKI0PC9R:~$ cd Content  
sreelekha@LAPTOP-CKI0PC9R:~/Content$ touch y  
sreelekha@LAPTOP-CKI0PC9R:~/Content$ ls  
x y  
sreelekha@LAPTOP-CKI0PC9R:~/Content$ mkdir -p dir1/dir2/dir3  
sreelekha@LAPTOP-CKI0PC9R:~/Content$ rm -r Content  
rm: cannot remove 'Content': No such file or directory  
sreelekha@LAPTOP-CKI0PC9R:~/Content$ rm -r  
rm: missing operand  
Try 'rm --help' for more information.  
sreelekha@LAPTOP-CKI0PC9R:~/Content$ cd ~  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -R Content  
Content:  
dir1 x y  
  
Content/dir1:  
dir2  
  
Content/dir1/dir2:  
dir3  
  
Content/dir1/dir2/dir3:  
sreelekha@LAPTOP-CKI0PC9R:~$ rm -r Content
```

Machine Test 03 – File Permissions

Objectives

1. Create a file called **test.txt**.
2. Display the current permissions of **test.txt**.
3. Change the permissions of **test.txt** so that:
 - Owner has **read and write** permissions.
 - Group has **read-only** permission.
 - Others have **read-only** permission.
4. Verify the updated permissions of **test.txt**.
5. Create a shell script called **deploy.sh**.
6. Grant executable permission to **deploy.sh**.
7. Verify that **deploy.sh** is executable.
8. Create a directory called **secure_data**.
9. Assign the following permissions to **secure_data**:
 - Owner has **full access**.
 - Group has **read and execute** permissions.
 - Others have **no permissions**.
10. Verify the permissions of **secure_data**.
11. Remove **test.txt**, **deploy.sh**, and **secure_data**.

Output:----

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ touch test.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -l test.txt  
-rw-r--r-- 1 sreelekha sreelekha 0 Oct  3 09:38 test.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ chmod 644 test.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -l test.txt  
-rw-r--r-- 1 sreelekha sreelekha 0 Oct  3 09:38 test.txt  
sreelekha@LAPTOP-CKI0PC9R:~$ touch deploy.sh  
sreelekha@LAPTOP-CKI0PC9R:~$ echo "Deployment Started" > deploy.sh  
sreelekha@LAPTOP-CKI0PC9R:~$ chmod +x deploy.sh  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -l deploy.sh  
-rwxr-xr-x 1 sreelekha sreelekha 19 Oct  3 09:43 deploy.sh  
sreelekha@LAPTOP-CKI0PC9R:~$ mkdir secure_data  
sreelekha@LAPTOP-CKI0PC9R:~$ chmod 750 secure_data  
sreelekha@LAPTOP-CKI0PC9R:~$ ls -ld secure_data  
drwxr-x--- 2 sreelekha sreelekha 4096 Oct  3 09:45 secure_data  
sreelekha@LAPTOP-CKI0PC9R:~$ rm test.txt deploy.sh  
sreelekha@LAPTOP-CKI0PC9R:~$ rmdir secure_data  
sreelekha@LAPTOP-CKI0PC9R:~$ ls  
AWS          Linux          backup         linux-test  
DevOps       LinuxPractice  config.env    project  
DevOps-Team  MyLinuxLab    content       project.txt  
DevOpsLab    Test          devops  
Docker       Ubuntu        employee.txt  
Kubernetes   application.log linux
```

Machine Test 04 – Users and Groups

Objectives

1. Create a new user called **devopsuser**.
2. Verify that the user **devopsuser** has been created successfully.
3. Create a new group called **devops**.
4. Add **devopsuser** to the **devops** group.
5. Verify the group membership of **devopsuser**.
6. Create a directory called **project_data**.
7. Change the ownership of **project_data** to **devopsuser**.
8. Change the group ownership of **project_data** to **devops**.
9. Verify the ownership and permissions of **project_data**.
10. Create a file called **project.txt** inside **project_data**.
11. Assign ownership of **project.txt** to **devopsuser** and **devops** group.
12. Verify the ownership details of **project.txt**.
13. Delete the **project.txt** file.
14. Remove the **project_data** directory.
15. Delete the **devopsuser** user.
16. Delete the **devops** group.

Output:----

```
sreelekh@LAPTOP-CKI0PC9R: ~  
sreelekh@LAPTOP-CKI0PC9R:~$ sudo useradd devopsuser  
[sudo: authenticate] Password:  
useradd: user 'devopsuser' already exists  
sreelekh@LAPTOP-CKI0PC9R:~$ sudo useradd -m devopsuser  
useradd: user 'devopsuser' already exists  
sreelekh@LAPTOP-CKI0PC9R:~$ id devopsuser  
uid=1001(devopsuser) gid=1001(devopsuser) groups=1001(devopsuser)  
sreelekh@LAPTOP-CKI0PC9R:~$ sudo groupadd devops  
groupadd: group 'devops' already exists  
sreelekh@LAPTOP-CKI0PC9R:~$ sudo usermod aG devops devopsuser  
Usage: usermod [options] LOGIN  
  
Options:  
-a, --append                append the user to the supplemental GROUPS  
                             mentioned by the -G option without removing  
                             the user from other groups  
-b, --badname                allow bad names (DEPRECATED)  
-c, --comment COMMENT       new value of the GECOS field  
-d, --home HOME_DIR         new home directory for the user account  
-e, --expiredate EXPIRE_DATE set account expiration date to EXPIRE_DATE  
-f, --inactive INACTIVE     set password inactive after expiration  
                             to INACTIVE  
-g, --gid GROUP              force use GROUP as new primary group  
-G, --groups GROUPS          new list of supplementary GROUPS  
-h, --help                   display this help message and exit  
-l, --login NEW_LOGIN        new value of the login name  
-L, --lock                   lock the user account  
-m, --move-home              move contents of the home directory to the  
                             new location (use only with -d)  
-o, --non-unique              allow using duplicate (non-unique) UID  
-p, --password PASSWORD      use encrypted password for the new password -P, --prefix PREFIX_DIR prefix directory where are located the /etc/* files  
-r, --remove                  remove the user from only the supplemental GROUPS  
                             mentioned by the -G option without removing  
                             the user from other groups  
-R, --root CHROOT_DIR        directory to chroot into  
-s, --shell SHELL            new login shell for the user account  
-u, --uid UID                 new UID for the user account  
-U, --unlock                  unlock the user account  
-v, --add-subuids FIRST-LAST add range of subordinate uids  
-V, --del-subuids FIRST-LAST remove range of subordinate uids  
-w, --add-subgids FIRST-LAST add range of subordinate gids  
-W, --del-subgids FIRST-LAST remove range of subordinate gids
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ groups devopsuser
devopsuser : devopsuser
sreelekha@LAPTOP-CKI0PC9R:~$ mkdir project_data
mkdir: project_data: File exists
sreelekha@LAPTOP-CKI0PC9R:~$ sudo chown devopsuser:devops project_data
sreelekha@LAPTOP-CKI0PC9R:~$ ls -ld project_data
drwxr-xr-x 2 devopsuser devops 4096 Oct  3 10:12 project_data
sreelekha@LAPTOP-CKI0PC9R:~$ sudo touch project_data/project.txt
sreelekha@LAPTOP-CKI0PC9R:~$ sudo chown devopsuser:devops project_data/project.txt
sreelekha@LAPTOP-CKI0PC9R:~$ ls -l project_data/project.txt
-rw-r--r-- 1 devopsuser devops 0 Oct  3 10:20 project_data/project.txt
sreelekha@LAPTOP-CKI0PC9R:~$ rm project_data/project.txt
rm: remove write-protected regular empty file 'project_data/project.txt'? yrm: cannot remove 'project_data/project.txt': Permission denied
sreelekha@LAPTOP-CKI0PC9R:~$ sudo rm project_data/project.txt
sreelekha@LAPTOP-CKI0PC9R:~$ sudo rmdir project_data
sreelekha@LAPTOP-CKI0PC9R:~$ sudo userdel -r devopsuser
userdel: devopsuser mail spool (/var/mail/devopsuser) not found
userdel: devopsuser home directory (/home/devopsuser) not found
sreelekha@LAPTOP-CKI0PC9R:~$ sudo groupdel devops
sreelekha@LAPTOP-CKI0PC9R:~$
```